

Part A - Conceptual Understanding (Theory)

Answers

1. What is Logistic Regression and why is it suitable for classification?

Logistic Regression is a supervised machine learning algorithm used for classification problems. It predicts the probability that a data point belongs to a particular class, making it suitable for binary classification tasks such as Risk Alert Prediction.

2. Explain classification performance metrics and why accuracy alone is insufficient.

Common metrics include Accuracy, Precision, Recall, F1-Score, ROC-AUC and Confusion Matrix. Accuracy alone can be misleading for imbalanced datasets because a model may predict the majority class correctly while failing to identify the minority class.

3. Define Type-I Error and Type-II Error in the context of risk prediction.

Type-I Error (False Positive) means a safe case is predicted as risky. Type-II Error (False Negative) means a risky case is predicted as safe. In risk prediction, Type-II Error is often more serious because actual risky cases may be missed.

4. Explain Precision, Recall, F1-Score, TPR, and FPR.

Precision measures how many predicted positive cases are actually positive. Recall (TPR) measures how many actual positive cases are correctly identified. F1-Score is the balance between Precision and Recall. FPR is the proportion of negative cases incorrectly classified as positive.

5. What is AUC-ROC and how does it help in evaluating classifiers?

AUC-ROC evaluates a classifier's ability to distinguish between classes across different thresholds. A higher AUC value indicates better classification performance.

6. Why does imbalanced data create problems in classification models?

In imbalanced datasets, one class has far more samples than the other. Models become biased toward the majority class, reducing their ability to detect minority-class cases accurately.